A NOVEL AND EFFICIENT SOLUTION TO BLOCK-BASED JOINT-DETECTION USING APPROXIMATE CHOLESKY FACTORIZATION

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Abstract *- Despite the fact that the zero-forcing block-linear equalizer (ZF-BLE) represents an entry-level solution to the joint-detection problem as applied to the TD-CDMA air-interface proposal for UMTS, it is nevertheless accompanied by a prohibitive computational complexity when supporting long data sequences, multiple users and channels with long impulse responses. A primary contributor to this computational complexity is the need to perform a Cholesky factorization of a sparse yet large correlation matrix. This paper presents a novel technique which exploits the pseudo block-Toeplitz nature of the Cholesky factor to derive an approximate triangular factorization, thereby allowing significant reductions in computational complexity and finite-precision effects at the expense of little or no degradation in performance. The technique is equally applicable to other block-based linear or decision-feedback joint-detection schemes whose operation relies on the Cholesky factorization.*

I. INTRODUCTION

As a result of recent standardization activities for the specification of an air-interface for third generation cellular systems in Europe, the mobile communications industry has witnessed the emergence of two key multiple-access proposals for UMTS, namely Wideband CDMA and TD-CDMA [1]. This paper presents an efficient technique for the implementation of detection and equalization in the latter technology.

The TD-CDMA multiple-access scheme represents an attempt to combine the advantages of continuous-time CDMA systems (such as frequency/interference diversity and soft capacity) with those of TDMA systems (such as simplified transceiver design and immunity against intra-cell interference) [2]. A central feature of the TD-CDMA concept is the simultaneous transmission and reception of multiple direct sequence spread signals within a time-slot and at the same carrier frequency. The TDMA structure of the proposal is identical to that of GSM, with ×8 time-slots contained within a frame of 60/13 ms duration. Each spread signal-burst within a time-slot corresponds to two blocks of N data symbols separated by a midamble. Synchronization occurs naturally on the downlink and is achieved via timing advance on the uplink. Extraction of the user data is subsequently performed at the receiver through the use of near-far resistant coherent joint-detection schemes, this being a requirement at both the mobile and base terminals. The use of multi-user joint-detection is an attempt to overcome the well-known near-far effects associated with single-user schemes commonly deployed in today's CDMA systems [3].

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Signal propagation through the radio channel results in both inter-symbol interference (ISI) and multiple-access interference (MAI) in the TD-CDMA system. The ISI mechanism is identical to that observed in typical TDMA wireless systems and refers to the filtering effect of the composite channel which forces data symbols within each of the individual spread signals to interfere with their respective neighbouring symbols. Filtering by the composite channel also implies that the spreading codes are no longer orthogonal at the input to the detector, even if they were designed to be mutually orthogonal at the transmitter. As a result of this non-orthogonality, symbols of the different spread signals will interfere with one another, each acting as a source of noise toward others. The joint-detection process attempts to make estimates of all transmitted user symbols by suppressing the combined effects of ISI and MAI.

A number of block-based joint-detection schemes which can be directly deployed in a TD-CDMA receiver have been investigated in recent publications [4]. The Zero-Forcing Block Linear Equalizer (ZF-BLE) represents the most basic of these schemes and essentially involves the solution of a linear Least-Squares problem. The ZF-BLE can be most efficiently implemented through the use of Cholesky factorization followed by forward and backward substitutions [5][6]. However, given the parameters of the TD-CDMA air-interface and the amount of delay spread observed in typical cellular propagation environments, the computational complexity associated with the ZF-BLE can rapidly exceed the capabilities of current digital signal processing technology.

The technique presented in this paper involves the use of an approximate Cholesky factorization which significantly reduces the computational complexity associated with the ZF-BLE, at the expense of no noticeable degradation in the detection performance.

The organization of the paper is as follows: In Section (II) the system model and the ZF-BLE are described. Section (III) contains a description of the standard Cholesky approach for the solution of the ZF-BLE. In Section (IV) details of the approximate Cholesky factorization are presented and the associated computational complexity is evaluated. Section (V) contains simulation results. Notation: vectors are underlined lower-case and matrices are in upper-case bold face. $\boldsymbol{A} \in C^{n \times m}$ signifies an $n \times m$ complex matrix, $\underline{a} \in C^n$ signifies an n-element complex column vector, $[\boldsymbol{A}]_{mn}$ signifies the element on the mth row and nth column of $\boldsymbol{A}$, while $(.)^*$, $(.)^H$ and $|.|$ indicate complex conjugate, complex conjugate transpose and vector 2-norm respectively. Although the results of this paper apply to both single-code and multi-code user communication, for purposes of clarity, a "user" is associated with a single spreading code without loss of generality.

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II. SYSTEM MODEL AND THE ZF-BLE

A block transmission TD-CDMA system is considered [4] in which K user signals occupy the same channel at the same time and in the same frequency band. Figure (1) illustrates the complex baseband model for signal generation and detection in such a system. Each kth user transmits a data symbol sequence $\underline{x}^{(k)}$ consisting of N symbols at symbol interval T_s , derived from the user data through a process of linear modulation. The total sequence of NK transmitted data symbols may be denoted by a vector $\underline{x} \in C^{NK}$ whose $[(n-1)K+k]^{\text{th}}$ element is the nth symbol of the kth user signal (n=1…N and k=1…K). Each data symbol of the kth user is multiplied by a user-specific signature sequence or code $\underline{c}^{(k)} \in C^{Q}$ consisting of Q (in general) complex chips at chip interval $T_c = T_s/Q$. The K codes are designed to be mutually orthogonal. Each of the K user signals is then transmitted over a composite radio channel (including Tx/Rx filters and air-interface) characterized by an impulse response $\underline{h}^{(k)} \in C^{W}$ of W samples at intervals T_c . Both ISI and MAI arise for W greater than unity. The received sequence $\underline{b} \in C^{NQ+W-1}$ (sampled at chip rate) consists of the sum of K individual sequences, each of length $NQ+W-1$, which arrive within a time-slot of the TD-CDMA air-interface. The received sequence may then be written as

$$\boldsymbol{A}\underline{x} + \underline{n} = \underline{b} \tag{1}$$

where $\boldsymbol{A} \in C^{(NQ+W-1)\times NK}$ is the channel/code convolution matrix and $\underline{n} \in C^{NQ+W-1}$ is additive stationary noise with zero mean and covariance matrix $\boldsymbol{R}_{nn} = \mathrm{E}\{\underline{n}\underline{n}^{H}\}$. The additive noise is usually assumed to be white for practical purposes, in which case $\boldsymbol{R}_{nn} = \sigma^2 \boldsymbol{I}$ where σ^2 is the noise variance. It can be readily seen that the convolution matrix $\boldsymbol{A}$ has a block-banded structure and consists of N replicas of a non-sparse submatrix $\boldsymbol{B} \in C^{(Q+W-1)\times K}$ arranged such that

$$[\boldsymbol{A}]_{(n-1)Q+1\,:\,(n-1)Q+V,\,(n-1)K+1\,:\,nK} = \boldsymbol{B} \tag{2}$$

where $V = Q+W-1$ and n=1…N. The kth column of the submatrix $\boldsymbol{B}$ is a vector which represents the convolution of the kth channel impulse response $\underline{h}^{(k)}$ (W chips) with the kth code $\underline{c}^{(k)}$ (Q chips), i.e.

$$[\boldsymbol{B}]_{1\,:\,V,\,k} = \underline{h}^{(k)} * \underline{c}^{(k)} \tag{3}$$

Given that matrix $\boldsymbol{A}$ is block-banded with a total of $NK(W+Q-1)$ non-zero elements, it can be shown that the Hermitian product $\boldsymbol{A}^H\boldsymbol{A}$ is block-banded, block multi-diagonal, and block-Toeplitz. In general, the product $\boldsymbol{A}^H\boldsymbol{A}$ has a block $(2\rho+1)$-diagonal structure where ρ depends on the degree of overlap between the non-zero submatrices of $\boldsymbol{A}$. Specifically, parameter ρ depends on the relationship between the length, W, of the channel impulse response and the number, Q, of chips per symbol and is equal to the largest integer for which the inequality $(\rho-1)Q+2 \le W$ is valid. In the example of Section (IV), the Hermitian product is block-tridiagonal (ρ=1), i.e. $2 \le W < Q+2$. Values of K=1…8, Q=16, $N \in \{28, 34\}$, $W \in \{12, 33, 34, 57, 117\}$ and a chip-rate of 13/6=2.1667 MHz have been adopted for the TD-CDMA proposal [1].

The objective of the receiver is to estimate $\underline{x}$ from the received signal $\underline{b}$ given an estimate, $\hat{\boldsymbol{A}}$, of the channel/code convolution matrix $\boldsymbol{A}$. The structure of $\hat{\boldsymbol{A}}$ is identical to that described in Equations (2) and (3) but with the channel impulse response $\underline{h}^{(k)}$ replaced by its estimated value. Midamble-assisted estimates of $\boldsymbol{A}$ may be derived through any one of several known techniques [7] and will not be considered further in this paper (a correlation-based algorithm for the estimation of $\boldsymbol{A}$ and timing acquisition is used in the simulations of Section V). It can be shown that, under the assumption of additive white Gaussian noise $\underline{n}$, the Maximum Likelihood estimate of $\underline{x}$ is given by the solution to the following classical Least-Squares problem:

$$\underline{\hat{x}}_{ML} = \min_{\underline{x}} \left| \hat{\boldsymbol{A}}\,\underline{x} - \underline{b} \right|_2^2 \tag{4}$$

Solution of above problem using an efficient dynamic programming search algorithm (e.g. Viterbi) which incorporates the appropriate constraints on the symbols x_i i=1…NK within the search process can indeed provide an optimal Maximum Likelihood Sequence Estimator (MLSE). Such an approach, however, would be computationally infeasible due to the large number of resulting trellis states. The ZF-BLE circumvents this problem by first performing a one-shot estimate of the sequence $\underline{x}$ and then incorporating the constraints on the symbols via a quantization process. This corresponds to an algebraic solution of the Least-Squares problem and represents a sub-optimal but computationally feasible detection scheme.
The estimate $\underline{\hat{x}}$ is given by the solution to the normal equation:

$$\left(\hat{\boldsymbol{A}}^H\hat{\boldsymbol{A}}\right)\underline{\hat{x}} = \hat{\boldsymbol{A}}^H\underline{b} \quad \text{where} \quad \boldsymbol{X} = \hat{\boldsymbol{A}}^H\hat{\boldsymbol{A}} \tag{5}$$

III. THE CHOLESKY TECHNIQUE FOR THE IMPLEMENTATION OF THE ZF-BLE

A large number of techniques exist for the solution of a linear system of equations [5][6]. These include brute-force matrix inversion, elimination algorithms (e.g. Gauss-Jordan or Gaussian elimination), triangular factorization (e.g. LU, LDM^H, LDL^H, Cholesky LL^H), and orthogonal factorization (e.g. QR). However, given that $\boldsymbol{X}$ is a Hermitian matrix, the most efficient technique for the solution of the normal equation involves the use of the Cholesky factorization followed by forward/backward substitution as described below:

STEP* (1) - *Cholesky Factorization:

The Hermitian matrix $\boldsymbol{X} = \hat{\boldsymbol{A}}^H\hat{\boldsymbol{A}} \in C^{NK\times NK}$ is factorized into the product of a lower triangular matrix $\boldsymbol{L} \in C^{NK\times NK}$ and its upper-triangular conjugate transpose $\boldsymbol{L}^H$, i.e. :

$$\boldsymbol{X} = \hat{\boldsymbol{A}}^H\hat{\boldsymbol{A}} = \boldsymbol{L}\boldsymbol{L}^H \qquad \text{so that} \qquad \underbrace{\boldsymbol{L}\boldsymbol{L}^H\,\underline{\hat{x}}}_{\underline{y}} = \hat{\boldsymbol{A}}^H\underline{b} = \underline{d} \tag{6}$$

Cholesky factorization is performed as follows where $\alpha_{ij} = [\hat{\boldsymbol{A}}^H\hat{\boldsymbol{A}}]_{ij}$:

for $j = 1 : NK$

$$[\boldsymbol{L}]_{jj} = \sqrt{\alpha_{jj} - \sum_{k=1}^{j-1} [\boldsymbol{L}]_{jk}[\boldsymbol{L}]_{jk}^*}$$

for $i = j+1 : NK$ (7)

$$[\boldsymbol{L}]_{ij} = \frac{1}{[\boldsymbol{L}]_{jj}^*}\left\{\alpha_{ij} - \sum_{k=1}^{j-1} [\boldsymbol{L}]_{ik}[\boldsymbol{L}]_{jk}^*\right\}$$

end

end

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STEP (2) - Solve for $\underline{y}$ using Forward Substitution:

Solve the lower-triangular system: $L\,\underline{y} = \underline{d}$ (8)

Since L is lower-triangular, it is possible to compute $\underline{y}$ efficiently as follows:

$$\underline{y}_1 = \underline{d}_1 / [L]_{11}$$

for $i = 2 : NK$ (9)

$$\underline{y}_i = \frac{1}{[L]_{ii}} \left\{ \underline{b}_i - \sum_{j=1}^{i-1} [L]_{ij}\, \underline{y}_j \right\}$$

end

STEP (3) - Solve for $\hat{\underline{x}}$ using Backward Substitution:

Solve the upper-triangular system: $L^H \hat{\underline{x}} = \underline{y}$ (10)

Since L^H is upper-triangular, it is possible to compute $\hat{\underline{x}}$ efficiently as follows:

$$\hat{\underline{x}}_{NK} = \underline{y}_{NK} / [L^H]_{NK \times NK}$$

for $i = NK-1 : -1 : 1$ (11)

$$\hat{\underline{x}}_i = \frac{1}{[L^H]_{ii}} \left\{ \underline{y}_i - \sum_{j=i+1}^{NK} [L^H]_{ij}\, \hat{\underline{x}}_j \right\}$$

end

Despite the fact that the ZF-BLE represents the most basic of joint-detection schemes, it can still be computationally expensive when supporting long data sequences (N>>1), large numbers of user signals (K>>1) and channels with long impulse responses (W>>1). A primary contributor to this is the need to perform a Cholesky factorization of the $NK \times NK$ correlation matrix $\boldsymbol{X}$, which is associated with a complexity of $O[(NK)^3/6]$ floating point operations (flops). For typical TD-CDMA parameters of N=34 and K=8, this alone can readily approach 10^9 floating point operations per second (Gflops/s). While the sparse nature of the correlation matrix $\boldsymbol{X}$ may be exploited to reduce this complexity down to tens of Mflops/s, the factorization still represents a significant load for current digital signal processing technology.

Table (1) presents the complexity requirements of the Cholesky factorization technique for the implementation of the ZF-BLE. It has been assumed that the sparse natures of matrices $\boldsymbol{A}$ and $\boldsymbol{X}$ are fully exploited. The sparsity of $\boldsymbol{A}$ defines the complexities of matched filtering and the generation of matrix $\boldsymbol{X}$. The sparsity of $\boldsymbol{X}$ affects the complexity of all other operations and is reflected in the parameter m which signifies the "average bandwidth" of the Cholesky factor $\boldsymbol{L}$. The operation-counts involving m are upper-bounds since end-effects have not been accounted for. The bounds are close to the actual values so long as NK>>m, which is the case for TD-CDMA. Furthermore, operations (*ii*), (*v*) and (*vi*) need to be performed twice for each time-slot (once for each data block). This has been accounted for by the ×2 multiplier.

The number of mathematical operations in Table (1) may be translated into the number of necessary floating-point DSP instructions per second according to the following rules: A complex multiply-and-add is treated as ×4 real multiply-and-adds. A real multiply-and-add or multiply is assumed to require one DSP cycle or instruction. A real addition or subtraction is assumed to require one DSP cycle or instruction. A real division is assumed to require ×6 DSP cycles or instructions. A square root is assumed to require ×10 DSP cycles or instructions. The latter two instruction-counts are typical of several commercial floating-point DSP products. Allowed time for real-time processing of a single time-slot is one TDMA frame period (60/13=4.6154 ms). This implies that for a time-slot containing K=8 user signals each corresponding to two symbols blocks of size N=28, with Q=16 chips per symbol and W=57 (dispersion of 26 μs for a chip-rate of 2.1667 MHz) or ρ=3, the ZF-BLE would require a total of approximately 217 MIPS of processing power, 143 MIPS of which is used exclusively to perform the Cholesky decomposition.

IV. THE APPROXIMATE CHOLESKY TECHNIQUE

The technique presented in this paper involves the use of an approximation which can significantly reduce the complexity associated with the Cholesky decomposition step of the ZF-BLE.

It is well known that any upper/lower triangular factor, $\boldsymbol{L}$, of a banded matrix, $\boldsymbol{X}$, maintains the upper/lower band structure of $\boldsymbol{X}$ [5]. It is also known that an upper/lower triangular factor, $\boldsymbol{L}$, of a Toeplitz matrix $\boldsymbol{X}$ does not maintain the Toeplitz structure of $\boldsymbol{X}$. These features also directly apply to the case of block-banded and block-Toeplitz structures. Consequently, despite the fact that the correlation matrix $\boldsymbol{X}$ is in general block $(2\rho+1)$-diagonal and block-Toeplitz, the resulting lower triangular Cholesky factor $\boldsymbol{L}$ is block-banded but not block-Toeplitz. This is illustrated in Equation (12) for the block tridiagonal case ρ=1, where N=5 is chosen as a convenient example and each submatrix has dimensions $K \times K$.

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{X}_{11} & \boldsymbol{X}_{21}^H & & & \\ \boldsymbol{X}_{21} & \boldsymbol{X}_{11} & \boldsymbol{X}_{21}^H & & \\ & \boldsymbol{X}_{21} & \boldsymbol{X}_{11} & \boldsymbol{X}_{21}^H & \\ & & \boldsymbol{X}_{21} & \boldsymbol{X}_{11} & \boldsymbol{X}_{21}^H \\ & & & \boldsymbol{X}_{21} & \boldsymbol{X}_{11} \end{bmatrix} = \boldsymbol{L}\boldsymbol{L}^H \quad \boldsymbol{L} = \begin{bmatrix} \boldsymbol{U}_1^H & & & & \\ \boldsymbol{F}_{11} & \boldsymbol{U}_2^H & & & \\ & \boldsymbol{F}_{12} & \boldsymbol{U}_3^H & & \\ & & \boldsymbol{F}_{13} & \boldsymbol{U}_4^H & \\ & & & \boldsymbol{F}_{14} & \boldsymbol{U}_5^H \end{bmatrix}$$

(12)

The upper-triangular submatrices $\boldsymbol{U}_i$ and non-sparse submatrices $\boldsymbol{F}_{1i}$ (i=1…N) of the Cholesky factor may be computed iteratively as follows:

$$\boldsymbol{U}_1 = chol[\boldsymbol{X}_{11}]$$
$$\boldsymbol{F}_{11} = \boldsymbol{X}_{21}\boldsymbol{U}_1^{-1}$$

for $i = 2 : N-1$

$$\boldsymbol{U}_i = chol[\boldsymbol{X}_{11} - \boldsymbol{F}_{1,i-1}\boldsymbol{F}_{1,i-1}^H] \quad (13)$$
$$\boldsymbol{F}_{1i} = \boldsymbol{X}_{21}\boldsymbol{U}_i^{-1}$$

end

$$\boldsymbol{U}_N = chol[\boldsymbol{X}_{11} - \boldsymbol{F}_{1,N-1}\boldsymbol{F}_{1,N-1}^H]$$

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As can be seen, the computational requirements consist of $\times N$ Cholesky factorizations, $\times(N-1)$ matrix-matrix Hermitian products, $\times(N-1)$ matrix by triangular-matrix products, $\times(N-1)$ triangular-matrix inversions and $\times(N-1)$ matrix subtractions. However, closer inspection of the Cholesky factor L reveals that the submatrices along each block diagonal are monotonically convergent in the sense that

$$\left(U_i - U_{i-1}\right) \to \varepsilon \qquad \left(F_{1,i} - F_{1,i-1}\right) \to \varepsilon \qquad \text{where } \varepsilon \to 0 \text{ as } i \to N \tag{14}$$

In other words, the Cholesky factor L is nearly block-Toeplitz and may be described to within an arbitrary accuracy by computing only the first $i_m < N$ submatrices $U_1 \ldots U_{im}$ and $F_{11} \ldots F_{1,im}$, thereafter setting $U_i = U_{im}$, $F_{1,i} = F_{1,im}$ for $i > i_m$ thus forming an approximate Cholesky factor $\tilde{L}$. Naturally the approximation becomes increasingly accurate for larger values of the *replication index* i_m.

Extensive tests under a variety of multipath fading conditions indicate that a replication index of i_m=2 provides a sufficiently accurate approximation and introduces no noticeable degradations in the performance of the ZF-BLE. The above "Toeplitzification" process implies that an approximate Cholesky factor $\tilde{L}$ may be derived according to the following four equations only:

$$\begin{aligned} U_1 &= chol\left[X_{11}\right] \\ F_{11} &= X_{21} U_1^{-1} \\ U_2 &= chol\left[X_{11} - F_{11} F_{11}^H\right] \\ F_{12} &= X_{21} U_2^{-1} \end{aligned} \tag{15}$$

$U_i = U_2$ and $F_{1,i} = F_{12}$ for $i > 2$. This is illustrated in Equation (16).

$$X = \begin{bmatrix} X_{11} & X_{21}^H & & & \\ X_{21} & X_{11} & X_{21}^H & & \\ & X_{21} & X_{11} & X_{21}^H & \\ & & X_{21} & X_{11} & X_{21}^H \\ & & & X_{21} & X_{11} \end{bmatrix} \cong \tilde{L}\tilde{L}^H \quad \tilde{L} = \begin{bmatrix} U_1^H & & & & \\ F_{11} & U_2^H & & & \\ & F_{12} & U_2^H & & \\ & & F_{12} & U_2^H & \\ & & & F_{12} & U_2^H \end{bmatrix} \tag{16}$$

The ZF-BLE process is then completed by using forward and back-substitution, as detailed previously, to solve the following modified version of the normal equation:

$$\underbrace{\tilde{L}\tilde{L}^H}_{\underline{y}} \underline{\hat{x}} = \hat{A}^H \underline{b} = \underline{d} \tag{17}$$

For clarity, the Toeplitzification process was initially presented for the simple block-tridiagonal case of ρ=1. The convergence property of the Cholesky submatrices have also been observed for block multi-diagonal matrices X. As a result, the Toeplitzification technique can be readily extended to block multi-diagonal cases (where $\rho > 1$ or $Q+2 \le W$) by replacing algorithm (15) with the more general version presented next.

Toeplitzification Algorithm - If the Cholesky factor, L, of matrix X is written in terms of its constituent submatrices

$$\begin{aligned} U_i &= L_{(i-1)K+1:iK,\,(i-1)K+1:iK} \qquad \forall i = 1 \cdots N \\ F_{pi} &= L_{[p+(i-1)]K+1:(p+i)K,\,(i-1)K+1:iK} \qquad \forall i = 1 \cdots N\text{-}p \quad \forall p = 1 \cdots \rho \end{aligned} \tag{18}$$

then the approximate Cholesky factor $\tilde{L}$ may be derived as

$$\begin{aligned} \tilde{L}_{1:K,\,1:K} &= U_1 \\ \tilde{L}_{(i-1)K+1:iK,\,(i-1)K+1:iK} &= U_2 \quad \forall i = 2 \cdots N \end{aligned} \tag{19}$$

$$\begin{aligned} \tilde{L}_{pK+1:(p+1)K,\,1:K} &= F_{p1} \quad \forall p = 1 \cdots \rho \\ \tilde{L}_{[p+(i-1)]K+1:(p+i)K,\,(i-1)K+1:iK} &= F_{p2} \quad \forall p = 1 \cdots \rho \quad \forall i = 1 \cdots N\text{-}p \end{aligned} \tag{20}$$

where

$$\begin{aligned} U_1 &= chol(X_{11}) \\ F_{p1} &= X_{p+1,1} U_1^{-1} \quad p = 1 \cdots \rho \\ U_2 &= chol\left(X_{11} - F_{11} F_{11}^H\right) \\ F_{p2} &= \left(X_{p+1,1} - F_{p+1,1} F_{11}^H\right) U_2^{-1} \quad p = 1 \cdots \rho - 1 \\ F_{\rho 2} &= X_{\rho+1,1}\, U_2^{-1} \end{aligned} \tag{21}$$

and all matrices in Equation (21) are of dimensions $K \times K$.

The reduced number of matrix operations and the small dimensionality of the matrices involved implies a reduction in finite-precision arithmetic effects as compared to the exact Cholesky technique. More importantly, this technique provides significant savings in computational complexity as compared to the exact Cholesky as illustrated in Table (2) for a replication factor of i_m=2.

It can be seen from the expressions of Table (2) and for parameter values similar to those presented in Section (III), that the ZF-BLE using an approximate Cholesky factorization would require a total of about 78 MIPS of processing power, only 4.4 MIPS of which is used for the Cholesky step. This represents a significant computational saving factor of ×32 in the factorization step as compared to the exact approach.

V. SIMULATION RESULTS

The down-link performance ($\underline{h}^{(k)} = \underline{h}\ \forall k$) of the ZF-BLE using both exact and approximate Cholesky (replication factors i_m=1 and 2) was investigated through simulations under static and multipath fading TU50/EQ50 channel conditions for the TD-CDMA air-interface (carrier frequency 2GHz). The static channel consists of three impulses equally spaced at 14 chip periods and of amplitudes (1+j), (1+j)/$\sqrt{2}$ and (1+j)/$\sqrt{2}$ respectively. Other simulation parameters include: K=4 users, W=33, Q=16 chips/symbol, data blocks of N=28 QAM4 modulated symbols, chip-rate of 2.1667 MHz. The spreading code chips are alternating real/imaginary and are based on Walsh codes multiplied by a pseudo-random sequence. X is block-pentadiagonal, i.e. ρ=2 (since $Q+2 < W < 2Q+2$). These parameter values correspond to case 1.1 in the

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TD-CDMA specifications [1]. The channel is simulated at ×2 chip-rate while channel-estimation and coherent joint-detection are both performed at chip-rate. Channel estimation is performed based on maximum-energy correlation windowing via a midamble of length 264 chips (=KW chosen to allow joint-estimation of a maximum of K=8 uplink channels [7]). Transmit pulse-shaping is with linearized GMSK (BT=0.3) truncated to five chip periods. Receiver decimation/channel-selection filtering consists of two third-order pole-zero Chebychev sections (cut-off at 0.8 MHz). Performance is averaged over 1000 time-slots. In Figures (2) and (3), the BER performances of the exact and approximate Cholesky techniques are presented. Note that in all cases, approximation using a replication index of i_m=2 is accompanied by literally no performance degradations. For the static case, however, the replication index of i_m=1 can be seen to imply too coarse an approximation. The static channel was judiciously chosen to illustrate this feature.

VI. CONCLUSION

A technique for the implementation of the ZF-BLE has been presented which utilises an approximate Cholesky factorization to achieve significant reductions in computational complexity, the extent of which depends on the amount of dispersion experienced by the transmitted radio signal and the number of symbols in the sequence being detected. It has been shown that for the TD-CDMA macro-cellular scenario involving a dispersion window of 26μs, a reduction factor of at least ×32 can be achieved in the factorization step. Furthermore, simulations with typical multipath fading environments indicate that the approximation has no observable effects on the detection performance of the ZF-BLE. The technique is equally applicable to other block-based linear or decision-feedback joint-detection schemes (e.g. the MMSE-BLE, ZF-BDFE and MMSE-BDFE) whose operation relies on the Cholesky factorization.

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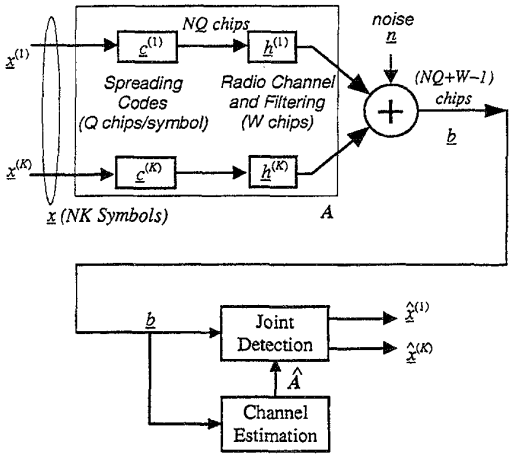


Figure (1) - Model for signal generation and detection.

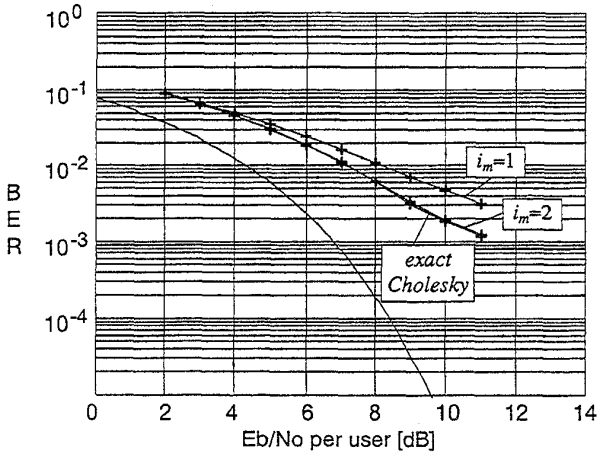


Figure (2) - Performance in a static multipath channel.

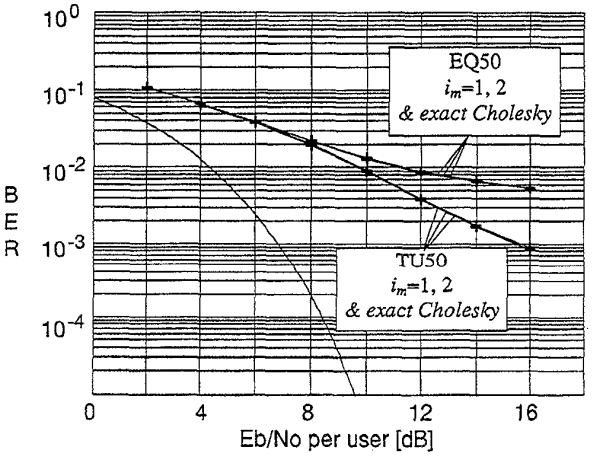


Figure (3) - Performance in fading multipath channels.

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TABLE (1) - Computational complexity of the exact Cholesky technique for the ZF-BLE.

	Description	Operation	Operation Count	
i	Matrix Generation	A	$K\,[W(Q-W+1)+W(W-1)]$ if $Q > W$	C-MACS
			$K\,[Q(W-Q+1)+Q(Q-1)]$ if $W > Q$	C-MACS
ii	Matched filtering	$\underline{d} = A^H \underline{b}$	$2 \times NK(Q+W-1)$	C-MACS
iii	Auto-correlation	$A^H A$	$(Q+W-1)K(K+1)/2 + K^2\rho\,[W-(\rho-1)Q/2-1]$	C-MACS
iv	Cholesky factor	$A^H A = LL^H$	$\leq NKm^2/2$	C-MACS
			$\leq NK(2m+1)$	R-SUBS
			$\leq 2NKm$	R-DIVS
			NK	R-SQRTS
v	Forward Substitution.	$L\underline{y} = \underline{d}$	$\leq 2 \times (NK-1)m$	C-MACS
			$2 \times 2(NK-1)$	R-SUBS
			$2 \times (2NK)$	R-DIVS
vi	Back Substitution	$L^H \underline{x} = \underline{y}$	$\leq 2 \times (NK-1)m$	C-MACS
			$2 \times 2(NK-1)$	R-SUBS
			$2 \times (2NK)$	R-DIVS

For Q=16, K=8, N=28, W=57 (ρ=3):

Total Complexity = 217 MIPS (per time-slot)
Complexity of Cholesky = 143 MIPS (per time-slot)

TABLE (2) - Computational complexity of the approximate Cholesky technique (i_m=2) for the ZF-BLE.

	Description	Operation	Operation Count	
i	Matrix Generation	A	$K\,[W(Q-W+1)+W(W-1)]$ if $Q > W$	C-MACS
			$K\,[Q(W-Q+1)+Q(Q-1)]$ if $W > Q$	C-MACS
ii	Matched filtering	$\underline{d} = A^H \underline{b}$	$2 \times NK(Q+W-1)$	C-MACS
iii	Auto-correlation	$A^H A$	$(Q+W-1)K(K+1)/2 + K^2\rho\,[W-(\rho-1)Q/2-1]$	C-MACS
iv	Cholesky factor	$A^H A \approx LL^H$	$(2\rho+2/3)K^3+\rho K^2-2K/3$	C-MACS
			$(2\rho+4)K^2-2K$	R-SUBS
			$2K^2-2K$	R-DIVS
			$2K$	R-SQRTS
v	Forward Substitution	$L\underline{y} = \underline{d}$	$\leq 2 \times (NK-1)m$	C-MACS
			$2 \times 2(NK-1)$	R-SUBS
			$2 \times (2NK)$	R-DIVS
vi	Backward Substitution	$L^H \underline{x} = \underline{y}$	$\leq 2 \times (NK-1)m$	C-MACS
			$2 \times 2(NK-1)$	R-SUBS
			$2 \times (2NK)$	R-DIVS

For Q=16, K=8, N=28, W=57 (ρ=3):

Total Complexity = 78 MIPS (per time-slot)
Complexity of approx. Cholesky = 4.4 MIPS (per time-slot)

where:

Q = Number of chips per symbol.
K = Number of users.
N = Number of symbols per block per user (2N per time-slot).
W = Length of channel impulse response in chip intervals.

$M = (\rho+0.5)K-1$ and

C-MACS = Complex multiply-and-accumulates
R-DIVS = Real divisions
R-SUBS = Real subtractions
R-SQRTS = Real square-roots

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